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Method Of Treating Major Depressive Disorder

Abstract

The present invention provides a method for safe and efficacious administration of esketamine.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation of U.S. Ser. No. 18/517,062, filed Nov. 22, 2023, which is a continuation of U.S. application Ser. No. 16/592,930, filed Oct. 4, 2019 (now U.S. Pat. No. 11,865,088), which claims the benefit of U.S. Provisional Application Nos. 62/741,562, filed Oct. 5, 2018 and 62/867,354, filed Jun. 27, 2019, and WO International Application PCT/IB2018/059107, filed Nov. 19, 2018, the entire contents of each of which are incorporated herein by reference.

FIELD OF THE INVENTION

[0002] The present disclosure provides methods for safe and efficacious administration of esketamine.

BACKGROUND OF THE INVENTION

[0003] Ketamine is a non-barbiturate, rapid acting, induction and general anesthetic agent that acts primarily via NMDA receptor antagonism in the CNS. The drug has been available in the United States since 1970 under the tradename Ketalar®. In 1971, DE2062620 described ketamine's (–) enantiomer, esketamine. Esketamine is not approved for use in the United States but is available in Europe as an induction and general anesthetic agent under the tradename Ketanest® S.

[0004] Sofia et al (1975) proposed the use of oral ketamine to treat depression. Berman et al (1980) described the results of a placebo controlled clinical trial of a single intravenous dose of ketamine in 7 patients with major depression. DE102007009888 suggests the use of (S)(+)-ketamine in the treatment of depression. More recently, there has been an increased interest in the possibility of using ketamine or esketamine for the treatment of major depressive disorder (MDD) including when the depression has proved refractory to other therapies.

[0005] Pharmaceutical compositions of ketamine and esketamine have been administered to healthy subjects and patients via a variety of routes of administration including intravenously, intranasally and orally. Clements et al (1982) record the relative bioavailability of oral ketamine as being 17% and of intramuscular ketamine as being 93%. Since that article, several other studies have recorded the relative oral bioavailability of ketamine as being between 17 and 24%.

Malinovsky et al (1996) record the relative bioavailability of intranasal ketamine as being 50% and of rectal ketamine as being 30%. Yanagihara et al (2003) record the relative bioavailability of both rectal and sublingual ketamine as being 30%, whereas they found nasal bioavailability to be 45%.

[0006] Although esketamine has been available for more than 40 years, there is very little published literature into its relative bioavailability by non-intravenous routes. Peltoniemi et al (2012) record the oral bioavailability of esketamine as being 11%, whereas Fanta et al (2015) found it to be only 8% and suggest that the first-pass metabolism of esketamine is more extensive than that found with ketamine. Unfortunately, although the protocol of a study into the relative bioavailability of intranasal and oral esketamine, NCT02343289, was already described in 2015, no results have been published. Daly et al (2017) record that 56 mg and 84 mg administered intranasally produces plasma esketamine levels that are in the pharmacokinetic range achieved by an intravenous administration of 0.2 mg/kg of esketamine, suggesting that the relative bioavailability of intranasal

esketamine might be considerably lower than that of ketamine.

[0007] The relative efficacy and safety of the two enantiomers of ketamine has also been a source of considerable debate in the literature. Ebert et al (1997) record that esketamine has a 5 times greater affinity for the NMDA receptor than (R)-ketamine. Oye et al (1992) record that esketamine was 4 times as potent as (R)-ketamine in reducing pain perception and in causing auditory and visual disturbances. Domino (2010) records that although esketamine appears more potent than (R)-ketamine, it also presents with greater undesirable psychotomimetic side effects. In contrast, Zhang et al (2014) and Yang et al (2015) have recorded that (R)-ketamine showed greater potency and longer-lasting antidepressant effects than esketamine in animal models of depression without psychotomimetic side effects and abuse liability. This has led some, such as Hashimoto (2016), to suggest that the anti-depressive effect of these molecules might not be due to NMDA receptor antagonism.

[0008] Despite the more recent interest in the use of ketamine and its enantiomers in the treatment of depression, most clinical reports describe the effects after a single administration. Blonk et al (2010) provide an extensive review of the doses recorded for chronic administration of oral ketamine in pain therapy and shows that typically high doses of 200 mg/d or more were prescribed for time periods of up to and greater than a year. Paslakis et al (2010) record four case reports of administering up to 1.25 mg/kg/d of oral esketamine as concomitant therapy in patients suffering from depression over a 14 day timeframe with two patients receiving up to 150 mg/d for 7 of their treatment days. Daly et al (2015) record the administration of 28 mg, 56 mg, and 84 mg of esketamine intranasally to patients twice a week for up to 14 days followed by an open label extension study of reduced dosing frequency for an additional 9 weeks, with all patients starting this extension on 56 mg/2.sup.nd week and most completing the study on 84 mg/2.sup.nd week.

[0009] Surprisingly, it has now been found that chronic administration of high levels of esketamine is associated with an increased risk of mutagenicity and therefore, circulating blood levels of the drug should be limited for administration of esketamine.

SUMMARY OF THE INVENTION

[0010] In one embodiment, the invention relates to a method of treating major depressive disorder (MDD) in a human patient in need thereof comprising orally administering to said patient an oral dosage form comprising between about 5 mg and about 40 mg of esketamine over a treatment regimen of at least 28 days.

Description

DETAILED DESCRIPTION OF THE INVENTION

[0011] The present invention is directed to methods of treating major depressive disorder in a human patient in need thereof comprising orally administering to said patient an oral dosage form comprising about 5 mg to about 40 mg (e.g., 5 mg to 40 mg) of esketamine over a treatment regimen of at least 28 days.

[0012] As used herein, the term “major depressive disorder”, or MDD, is characterized as a psychiatric disorder meeting five criteria: 1) the presence during the same 2 week period which together represent a change from previous functioning, of a depressed/sad mood or a loss of interest and pleasure, together with five (or more) of the following additional criteria occurring nearly every day i) depressed/sad mood ii) loss of interest and pleasure iii) significant weight loss when not dieting or weight gain or a decrease or increase in appetite iv) insomnia or hypersomnia v) psychomotor agitation or retardation vi) fatigue or loss of energy vii) feelings of worthlessness or excessive or inappropriate guilt viii) diminished ability to think or concentrate or indecisiveness ix) recurrent thoughts of death or suicidal ideation, planning or attempt: 2) the symptoms cause clinically significant distress or impairment in social, occupational or other functioning: 3) the

episode is not better accounted for by a psychotic disorder: 4) the episode is not attributable to the physiological effects of a substance or to another medical condition: 5) there has never been a manic or hypomanic episode (Diagnostic and Statistical Manual of Mental Disorders, 5th Edition, American Psychiatric Association, 2013). Other indications contemplated include treating, preventing, or ameliorating one or more symptoms of a disorder including, but not limited to, Rett syndrome, depression, refractory depression, suicidality, obsessive-compulsive disorder, fibromyalgia, post-traumatic stress syndrome, autism spectrum disorder, and depression associated with genetic disorders.

[0013] In one embodiment, the major depressive disorder is with anxious distress. In another embodiment, the disorder is with mixed features. In another embodiment, the disorder is with melancholic features. In another embodiment, the disorder is with atypical features. In another embodiment, the disorder is with mood-congruent psychotic features. In another embodiment, the disorder is with mood-incongruent psychotic features. In another embodiment, the disorder is with catatonia. In another embodiment, the disorder is with peripartum onset. In another embodiment, the disorder is with seasonal pattern.

[0014] In one embodiment, the major depressive disorder has not responded to adequate doses and treatment duration of antidepressants other than ketamine or esketamine. In some aspects, the non-responder has failed to demonstrate an improvement of up to 25% in MADRS score, or a similar psychometric score, after adequate doses and treatment duration of antidepressants other than ketamine or esketamine. In other aspects, the non-responder has demonstrated an incomplete improvement of between 25-50% in MADRS score, or a similar psychometric score, after adequate doses and treatment duration of antidepressants other than ketamine or esketamine. In other aspects, the non-responder has demonstrated an inadequate improvement of up to 50% in MADRS score, or a similar psychometric score, after adequate doses and treatment duration of antidepressants other than ketamine or esketamine. In some aspects, the adequate doses and treatment duration of antidepressants other than ketamine or esketamine, refers to doses and treatment duration of one, or more, antidepressants other than ketamine or esketamine during the current depressive episode. In other aspects, the adequate course refers to the non-response to doses and treatment duration of one, or more, antidepressants other than ketamine or esketamine during a previous depressive episode. In other aspects, the adequate course refers to the non-response to doses and treatment duration of one, or more, antidepressants other than ketamine or esketamine both during a previous depressive episode and during the current depressive episode. In some aspects, the disorder is treatment-refractory or treatment-resistant depression, i.e., depression that has failed to respond to adequate doses and treatment duration of at least two antidepressants other than ketamine or esketamine.

[0015] As used herein, the term “treating major depressive disorder” can refer to a reduction of the symptoms of Major Depressive Disorder, as measured by reduction in the Montgomery-Asberg Depression Rating Scale (MADRS) score. In some aspects, the term “treating major depressive disorder” refers to a change from baseline, as measured the MADRS score. In some aspects, the term “treating major depressive disorder” refers to a remission, as measured by reduction in the MADRS score. In some aspects, the term “treating major depressive disorder” refers to a 50% or greater improvement, as measured the MADRS score.

[0016] In other aspects, the term “treating major depressive disorder” refers to change from baseline on Sheehan Disability Scale (SDS).

[0017] In other aspects, the term “treating major depressive disorder” refers to change from baseline on self-rated Symptoms of Depression Questionnaire (SDQ).

[0018] In other aspects, the term “treating major depressive disorder” refers to change from baseline on the physician-administered Clinical Global Impression Improvement (CGI-I).

[0019] In other aspects, the term “treating major depressive disorder” refers to change from baseline on the physician-administered Global Impression Severity (SCG-S).

[0020] In other aspects, the term “treating major depressive disorder” refers to change from baseline on Generalized Anxiety Disorder 7 items scale.

[0021] The methods of the disclosure will exhibit an acceptable safety and/or tolerability profile. That is, the benefits achieved using the methods of the disclosure will outweigh any safety and/or tolerability considerations exhibited by using the disclosed methods, as compared to placebo. In other aspects, the benefits achieved using the methods of the disclosure will outweigh any safety and/or tolerability considerations exhibited by using the disclosed methods, as compared to other methods of treating MDD, including treatment-resistant MDD. Other methods of treating MDD, including treatment-resistant MDD include other methods of using ketamine and esketamine. For example, the benefits achieved using the methods of the disclosure will outweigh any adverse events including, for example, untoward changes in hematology, biochemistry, urinalysis, immunological parameters, physical examination findings, blood pressure, and/or heart rate, as compared to placebo. In other aspects, the benefits achieved using the methods of the disclosure will outweigh any adverse events including, for example, changes in hematology, biochemistry, urinalysis, immunological parameters, physical examination findings, blood pressure, and/or heart rate, as compared to other methods of treating MDD, including treatment-resistant MDD.

[0022] In other aspects, the benefits achieved using the methods of the disclosure with outweigh any adverse events in 12 lead ECG findings, method discontinuation, Digit Symbol Substitution Test (DSST), reaction time test (Cambridge COGNITION and/or Cogstate battery), self administered Stanford sleepiness scale, a Bladder Pain/Interstitial Cystitis Symptom Score (BPIC-SS), a Modified Observer's Alertness/Sedation Scale (MOAA/S), a Clinician-Administered Dissociative States Scale (CADSS), a Suicidality Scale-Clinician-Rated Columbia Suicide Severity Rating Scale (C-SSRS), 4 items positive symptoms subscale from the Brief Psychiatric Rating Scale (BPRS), and/or 20 item Physician Withdrawal Checklist (PWC-20), as compared to placebo. In other aspects, the benefits achieved using the methods of the disclosure with outweigh any adverse events in 12 lead ECG findings, method discontinuation, Digit Symbol Substitution Test (DSST), reaction time test (Cambridge COGNITION and/or Cogstate battery), self administered Stanford sleepiness scale, a Bladder Pain/Interstitial Cystitis Symptom Score (BPIC-SS), a Modified Observer's Alertness/Sedation Scale (MOAA/S), a Clinician-Administered Dissociative States Scale (CADSS), a Suicidality Scale-Clinician-Rated Columbia Suicide Severity Rating Scale (C-SSRS), 4 items positive symptoms subscale from the Brief Psychiatric Rating Scale (BPRS), and/or 20 item Physician Withdrawal Checklist (PWC-20), as compared to other methods of treating MDD, including treatment-resistant MDD.

[0023] As used herein, the term “ketamine” shall refer to the chemical compound dl 2-(2-chlorophenyl)-2(methylamino) cyclohexanone, or a pharmaceutically acceptable salt thereof.

[0024] As used herein, the term “esketamine” shall refer to the (S)-enantiomer of ketamine also known as the chemical compound (2S)-2-(2-Chlorophenyl)-2-(methylamino) cyclohexanone, or a pharmaceutically acceptable salt thereof. As used herein, the term “esketamine” shall be understood to be to the exclusion of the compound as found, without an enantiomeric excess, in ketamine, or a pharmaceutically acceptable salt thereof. In one embodiment, the esketamine, or a pharmaceutically acceptable salt thereof, is the hydrochloride salt of esketamine, i.e., esketamine hydrochloride.

[0025] As used herein, the term “(R)-ketamine” shall refer to the (R)-enantiomer of ketamine also known as the chemical compound (2R)-2-(2-Chlorophenyl)-2-(methylamino) cyclohexanone, or a pharmaceutically acceptable salt thereof. As used herein, the term “(R)-ketamine” shall be understood to be to the exclusion of the compound as found, without an enantiomeric excess, in ketamine, or a pharmaceutically acceptable salt thereof.

[0026] As used herein, the term “(S)-norketamine” shall refer to the (S)-enantiomer of norketamine also known as the chemical compound (2S)-2-(2-Chlorophenyl)-2-(amino) cyclohexanone, or a pharmaceutically acceptable salt thereof. As used herein, the term “(S)-norketamine” shall be

understood to be to the exclusion of the compound as found, without an enantiomeric excess, in norketamine, or a pharmaceutically acceptable salt thereof.

[0027] As used herein, the term “(2S,6S)-OH-norketamine” shall refer to the (2S,6S)-enantiomer of hydroxynorketamine also known as the chemical compound (2S,6S)-2-Amino-2-(2-chlorophenyl)-6-hydroxycyclohexanone, or a pharmaceutically acceptable salt thereof. As used herein, the term “(2S,6S)—OH-Norketamine” shall be understood to be to the exclusion of the compound as found, without an enantiomeric excess, in hydroxynorketamine, or a pharmaceutically acceptable salt thereof.

[0028] The chemical compounds described herein according to the invention are also intended to include such compounds wherein the molecular structures include isotopes of carbon, hydrogen and nitrogen atoms occurring on those structures. Isotopes include those atoms having the same atomic number but different mass numbers. For example, isotopes of hydrogen include deuterium. Isotopes of carbon include C-13. Isotopes of nitrogen include N-15.

[0029] Accordingly, within the chemical structure of any chemical compound taught in this application as suitable for the formulations disclosed herein: [0030] any hydrogen atom or group of hydrogen atoms, could suitably be replaced by an isotope of hydrogen, i.e., deuterium; [0031] any carbon atom or group of carbon atoms, could suitably be replaced by an isotope of carbon, i.e., ¹³C; and [0032] any nitrogen atom or group of nitrogen atoms, could suitably be replaced by an isotope of nitrogen, i.e., ¹⁵N.

[0033] As used herein, the term “treatment regimen” shall refer to time period during which the human patient, in need thereof, will be treated by more than one, either daily or intermittent, administrations of esketamine. In a preferred embodiment of the invention, the treatment regimen will extend for at least 28 days. In another preferred embodiment, the treatment regimen will extend for at least 30 days. In another preferred embodiment, the treatment regimen will be for 28 days to about 365 days. In another preferred embodiment, the treatment regimen will be for 28 days to about 730 days. Another preferred embodiment, the treatment regimen will extend for at least one month. In another preferred embodiment, the treatment regimen will extend for at least 1 year (365 days). In another preferred embodiment of the invention, the treatment regimen will extend for at least about 730 days, that is, at least about 2 years. In another embodiment, the treatment regimen varies over the course of the 28 to about 730 days (i.e., about two years). A medical professional skilled in the art of psychiatry will be able to determine the administration regimen over the 28 to about 730 days (e.g. about two years).

[0034] It is understood that where a parameter range is provided, all integers within that range, and tenths thereof, are also provided by the invention. For example, “1-30 ng/ml” includes 1.1 ng/ml, 1.2 ng/ml, 1.3 ng/ml, etc. up to 30 ng/ml. In another example, “0.1-2.5 mg/day” includes 0.1 mg/day, 0.2 mg/day, 0.3 mg/day, etc. up to 2.5 mg/day.

[0035] The present invention is further directed to a method of treating major depressive disorder in a human patient in need thereof by orally administering to said patient of an oral dosage form comprising about 5 mg to about 40 mg, preferably 5 mg to 40 mg, of esketamine over a treatment regimen of at least 28 days.

[0036] In some aspects, oral dosage forms of the disclosure include esketamine, i.e., esketamine as a free base. In other aspects, oral dosage forms of the disclosure include pharmaceutically acceptable salts of esketamine. As used herein, amounts of esketamine present in the oral dosage forms of the disclosure refer to amounts of esketamine free base. For example, in those aspects wherein the oral dosage form comprises esketamine free base, “10 mg of esketamine” refers to 10 mg of the esketamine free base in the oral dosage form. In aspects wherein the oral dosage form comprises a pharmaceutically acceptable salt of esketamine, such as esketamine hydrochloride, “10 mg of esketamine” refers to 10 mg esketamine free base, based on 11.53 mg of esketamine hydrochloride in the oral dosage form.

[0037] In one preferred embodiment of the invention, the oral administration to said patient is of an

oral dosage form comprising about 5 mg of esketamine, preferably 5 mg of esketamine. In another preferred embodiment of the invention, the oral administration to said patient is of an oral dosage form comprising about 10 mg of esketamine, preferably 10 mg of esketamine. In another preferred embodiment, the oral administration to said patient is of an oral dosage form comprising about 20 mg of esketamine, preferably 20 mg of esketamine. In another preferred embodiment, the oral administration to said patient is of an oral dosage form comprising about 30 mg of esketamine, preferably 30 mg of esketamine. In yet another preferred embodiment, the oral administration to said patient is of an oral dosage form comprising about 40 mg of esketamine, preferably 40 mg of esketamine.

[0038] Without wanting to be bound to any particular theory, it is believed that the therapeutic effect of repeated oral dosing of esketamine can be demonstrated by either the administration of higher doses of the drug at larger intervals or the administration of lower doses at shorter intervals. This allows for equivalent exposure over time of the drug and its metabolites and lower peak concentrations thereby reducing the overall risk of genotoxic events and improving the clinical safety profile.

[0039] The present invention is further directed to a method of treating major depressive disorder in a human patient in need thereof by the oral administration to said patient of an oral dosage form comprising about 5 mg to about 40 mg of esketamine, preferably 5 mg to 40 mg of esketamine, wherein the esketamine C.sub.max of said administration is 30 ng/ml or less.

[0040] As used herein, the term “C.sub.max” shall refer to the mean (average) observed maximum plasma concentration assayed after any single administration. In some embodiments the method disclosed herein further comprises measuring plasma levels in the patient.

[0041] In one embodiment of the invention, the esketamine C.sub.max of said administration is 30 ng/ml or less, 29 ng/ml or less, 28 ng/ml or less, 27 ng/ml or less, 26 ng/ml or less, 25 ng/ml or less, 24 ng/ml or less, 23 ng/ml or less, 22 ng/ml or less, 21 ng/ml or less, 20 ng/ml or less, 19 ng/ml or less, 18 ng/ml or less, 17 ng/ml or less, 16 ng/ml or less, 15 ng/ml or less, 14 ng/ml or less, 13 ng/ml or less, 12 ng/ml or less, 11 ng/ml or less, 10 ng/ml or less, 9 ng/ml or less, 8 ng/ml or less, 7 ng/ml or less, 6 ng/ml or less, 5 ng/ml or less, 4 ng/ml or less, 3 ng/ml or less, 2 ng/ml or less, or 1 ng/ml or less. In one preferred embodiment of the invention, the esketamine C.sub.max of said administration is 30 ng/ml or less. In another preferred embodiment of the invention, the esketamine C.sub.max of said administration is 15 ng/ml or less. In one preferred embodiment of the invention, the esketamine C.sub.max of said administration is between 15 ng/mL and 30 ng/mL. In one preferred embodiment of the invention, the esketamine C.sub.max of said administration is between 10 ng/mL and 15 ng/mL. In one preferred embodiment of the invention, the esketamine C.sub.max of said administration is between 5 ng/mL and 15 ng/mL. In one preferred embodiment of the invention, the esketamine C.sub.max of said administration is between 11 ng/mL and 13 ng/mL.

[0042] The present invention is further directed to a method of treating major depressive disorder in a human patient in need thereof by the oral administration to said patient of an oral dosage form comprising about 5 mg to about 40 mg of esketamine, for example, 5 mg to 40 mg of esketamine, wherein the esketamine AUC.sub.0-t of said administration is 60 ng*h/ml or less.

[0043] As used herein, the term “AUC” shall refer to the area under the plasma concentration/time curve after any single administration. The term “AUC.sub.0-t” shall refer to the area under the plasma concentration/time curve from time 0 to the last quantifiable concentration after any single administration and the term “AUC.sub.0-inf” shall refer to the area under the plasma concentration/time curve from time 0 until the extrapolated concentration at infinity after any single administration. The term “AUC.sub.tau” shall refer to the area under the plasma concentration/time curve over the steady state dosing interval.

[0044] In one embodiment of the invention, the esketamine AUC.sub.0-t of said administration is 60 ng*h/ml, 59 ng*h/ml, 58 ng*h/ml, 57 ng*h/ml, 56 ng*h/ml, 55 ng*h/ml, 54 ng*h/ml, 53

ng*h/ml, 52 ng*h/ml, 51 ng*h/ml, 50 ng*h/ml, 49 ng*h/ml, 48 ng*h/ml, 47 ng*h/ml, 46 ng*h/ml, 45 ng*h/ml, 44 ng*h/ml, 43 ng*h/ml, 42 ng*h/ml, 41 ng*h/ml, 40 ng*h/ml, 39 ng*h/ml, 38 ng*h/ml, 37 ng*h/ml, 36 ng*h/ml, 35 ng*h/ml, 34 ng*h/ml, 33 ng*h/ml, 32 ng*h/ml, 31 ng*h/ml, 30 ng*h/ml, 29 ng*h/ml, 28 ng*h/ml, 27 ng*h/ml, 26 ng*h/ml, 25 ng*h/ml, 24 ng*h/ml, 23 ng*h/ml, 22 ng*h/ml, 21 ng*h/ml, 20 ng*h/ml, 19 ng*h/ml, 18 ng*h/ml, 17 ng*h/ml, 16 ng*h/ml, 15 ng*h/ml, 14 ng*h/ml, 13 ng*h/ml, 12 ng*h/ml, 11 ng*h/ml, 10 ng*h/ml, 9 ng*h/ml, 8 ng*h/ml, 7 ng*h/ml, 6 ng*h/ml, 5 ng*h/ml, 4 ng*h/ml, 3 ng*h/ml, 2 ng*h/ml, or 1 ng*h/ml. In one preferred embodiment of the invention, the esketamine AUC.sub.0-t of said administration is 60 ng*h/ml or less. In another preferred embodiment of the invention, the esketamine AUC.sub.0-t of said administration is 30 ng*h/ml or less. In one preferred embodiment of the invention, the esketamine AUC.sub.0-t of said administration is between 30 ng*h/ml and 60 ng*h/ml. In one preferred embodiment of the invention, the esketamine AUC.sub.0-t of said administration is between 15 ng*h/ml and 30 ng*h/ml.

[0045] The present invention is further directed to a method of treating major depressive disorder in a human patient in need thereof by the oral administration to said patient of a dosage form, wherein said dosage form provides for an (S)-norketamine C.sub.max of 150 ng/ml or less.

[0046] In one embodiment of the invention, the (S)-norketamine C.sub.max of said administration is 150 ng/ml or less, 145 ng/ml or less, 140 ng/ml or less, 139 ng/ml or less, 138 ng/ml or less, 137 ng/ml or less, 136 ng/ml or less, 135 ng/ml or less, 134 ng/ml or less, 133 ng/ml or less, 132 ng/ml or less, 131 ng/ml or less, 130 ng/ml or less, 129 ng/ml or less, 128 ng/ml or less, 127 ng/ml or less, 126 ng/ml or less, 125 ng/ml or less, 120 ng/ml or less, 115 ng/ml or less, 110 ng/ml or less, 105 ng/ml or less, 100 ng/ml or less, 95 ng/ml or less, 90 ng/ml or less, 85 ng/ml or less, 80 ng/ml or less, 75 ng/ml or less, 74 ng/ml or less, 73 ng/ml or less, 72 ng/ml or less, 71 ng/ml or less, 70 ng/ml or less, 69 ng/ml or less, 68 ng/ml or less, 67 ng/ml or less, 66 ng/ml or less, 65 ng/ml or less, 64 ng/ml or less, 63 ng/ml or less, 62 ng/ml or less, 61 ng/ml or less, 60 ng/ml or less, 55 ng/ml or less, 50 ng/ml or less, 45 ng/ml or less, 40 ng/ml or less, 35 ng/ml or less, 34 ng/ml or less, 33 ng/ml or less, 32 ng/ml or less, 31 ng/ml or less, 30 ng/ml or less, 25 ng/ml or less, 20 ng/ml or less, 19 ng/ml or less, 18 ng/ml or less, 17 ng/ml or less, 16 ng/ml or less or 15 ng/ml or less. In one preferred embodiment of the invention, the (S)-norketamine C.sub.max of said administration is 150 ng/ml or less. In another preferred embodiment of the invention, the (S)-norketamine C.sub.max of said administration is 75 ng/ml or less. In another preferred embodiment of the invention, the (S)-norketamine C.sub.max of said administration is 35 ng/ml or less. In another preferred embodiment of the invention, the (S)-norketamine C.sub.max of said administration is 20 ng/ml or less. In one preferred embodiment of the invention, the (S)-norketamine C.sub.max of said administration is between 15 ng/mL and 150 ng/mL. In another preferred embodiment of the invention, the (S)-norketamine C.sub.max of said administration is between 15 ng/mL and 20 ng/mL. In another preferred embodiment of the invention, the (S)-norketamine C.sub.max of said administration is between 30 ng/mL and 35 ng/mL. In another preferred embodiment of the invention, the (S)-norketamine C.sub.max of said administration is between 60 ng/mL and 75 ng/mL. In another preferred embodiment of the invention, the (S)-norketamine C.sub.max of said administration is between 125 ng/mL and 140 ng/mL. In another preferred embodiment of the invention, the (S)-norketamine C.sub.max of said administration is between 60 ng/mL and 140 ng/mL.

[0047] The present invention is further directed to a method of treating major depressive disorder in a human patient in need thereof by the oral administration to said patient of a dosage form, wherein said dosage form provides for an (S)-norketamine AUC.sub.0-t of 850 ng*h/ml or less.

[0048] In one embodiment of the invention, the (S)-norketamine AUC.sub.0-t of said administration is 850 ng*h/ml or less, 845 ng*h/ml or less, 840 ng*h/ml or less, 839 ng*h/ml or less, 838 ng*h/ml or less, 837 ng*h/ml or less, 836 ng*h/ml or less, 835 ng*h/ml or less, 834 ng*h/ml or less, 832 ng*h/ml or less, 831 ng*h/ml or less, 830 ng*h/ml or less, 829 ng*h/ml or

less, 828 ng*h/ml or less, 827 ng*h/ml or less, 826 ng*h/ml or less, 825 ng*h/ml or less, 824 ng*h/ml or less, 823 ng*h/ml or less, 822 ng*h/ml or less, 821 ng*h/ml or less, 820 ng*h/ml or less, 815 ng*h/ml or less, 810 ng*h/ml or less, 805 ng*h/ml or less, 800 ng*h/ml or less, 795 ng*h/ml or less, 790 ng*h/ml or less, 785 ng*h/ml or less, 780 ng*h/ml or less, 775 ng*h/ml or less, 770 ng*h/ml or less, 765 ng*h/ml or less, 760 ng*h/ml or less, 755 ng*h/ml or less, 750 ng*h/ml or less, 745 ng*h/ml or less, 740 ng*h/ml or less, 735 ng*h/ml or less, 730 ng*h/ml or less, 725 ng*h/ml or less, 720 ng*h/ml or less, 710 ng*h/ml or less, 700 ng*h/ml or less, 690 ng*h/ml or less, 680 ng*h/ml or less, 670 ng*h/ml or less, 660 ng*h/ml or less, 650 ng*h/ml or less, 640 ng*h/ml or less, 630 ng*h/ml or less, 620 ng*h/ml or less, 610 ng*h/ml or less, 600 ng*h/ml or less, 590 ng*h/ml or less, 580 ng*h/ml or less, 570 ng*h/ml or less, 560 ng*h/ml or less, 550 ng*h/ml or less, 540 ng*h/ml or less, 530 ng*h/ml or less, 520 ng*h/ml or less, 510 ng*h/ml or less, 500 ng*h/ml or less, 490 ng*h/ml or less, 480 ng*h/ml or less, 470 ng*h/ml or less, 460 ng*h/ml or less, 450 ng*h/ml or less, 440 ng*h/ml or less, 430 ng*h/ml or less, 425 ng*h/ml or less, 420 ng*h/ml or less, 419 ng*h/ml or less, 418 ng*h/ml or less, 417 ng*h/ml or less, 416 ng*h/ml or less, 415 ng*h/ml or less, 414 ng*h/ml or less, 413 ng*h/ml or less, 412 ng*h/ml or less, 411 ng*h/ml or less, 410 ng*h/ml or less, 405 ng*h/ml or less, 400 ng*h/ml or less, 390 ng*h/ml or less, 380 ng*h/ml or less, 380 ng*h/ml or less, 370 ng*h/ml or less, 360 ng*h/ml or less, 350 ng*h/ml or less, 340 ng*h/ml or less, 330 ng*h/ml or less, 320 ng*h/ml or less, 310 ng*h/ml or less, 300 ng*h/ml or less, 290 ng*h/ml or less, 280 ng*h/ml or less, 270 ng*h/ml or less, 260 ng*h/ml or less, 250 ng*h/ml or less, 240 ng*h/ml or less, 230 ng*h/ml or less, 220 ng*h/ml or less, 215 ng*h/ml or less, 210 ng*h/ml or less, 209 ng*h/ml or less, 208 ng*h/ml or less, 207 ng*h/ml or less, 206 ng*h/ml or less, 205 ng*h/ml or less, 200 ng*h/ml or less, 190 ng*h/ml or less, 180 ng*h/ml or less, 170 ng*h/ml or less, 160 ng*h/ml or less, 150 ng*h/ml or less, 140 ng*h/ml or less, 130 ng*h/ml or less, 120 ng*h/ml or less, 110 ng*h/ml or less, 105 ng*h/ml or less, 104 ng*h/ml or less, 103 ng*h/ml or less or 102 ng*h/ml or less. In one preferred embodiment of the invention, the (S)-norketamine AUC.sub.0-t of said administration is 850 ng*h/ml or less. In another preferred embodiment of the invention, the (S)-norketamine AUC.sub.0-t of said administration is 420 ng*h/ml or less. In another preferred embodiment of the invention, the (S)-norketamine AUC.sub.0-t of said administration is 210 ng*h/ml or less. In another preferred embodiment of the invention, the (S)-norketamine AUC.sub.0-t of said administration is 105 ng*h/ml or less. In one preferred embodiment of the invention, the (S)-norketamine AUC.sub.0-t of said administration is between 105 ng*h/ml and 850 ng*h/ml. In another preferred embodiment of the invention, the (S)-norketamine AUC.sub.0-t of said administration is between 105 ng*h/ml and 850 ng*h/ml. In another preferred embodiment of the invention, the (S)-norketamine AUC.sub.0-t of said administration is between 102 ng*h/ml and 105 ng*h/ml. In another preferred embodiment of the invention, the (S)-norketamine AUC.sub.0-t of said administration is between 205 ng*h/ml and 210 ng*h/ml. In another preferred embodiment of the invention, the (S)-norketamine AUC.sub.0-t of said administration is between 410 ng*h/ml and 420 ng*h/ml. In another preferred embodiment of the invention, the (S)-norketamine AUC.sub.0-t of said administration is between 820 ng*h/ml and 840 ng*h/ml.

[0049] The present invention is further directed to a method of treating major depressive disorder in a human patient in need thereof by the oral administration to said patient of a dosage form, wherein said dosage form provides for an (2S,6S)-OH-norketamine C.sub.max of 75 ng/ml or less.

[0050] In one embodiment of the invention, the (2S,6S)-OH-norketamine C.sub.max of said administration is 75 ng/ml or less, 74 ng/ml or less, 73 ng/ml or less, 72 ng/ml or less, 71 ng/ml or less, 70 ng/ml or less, 69 ng/ml or less, 68 ng/ml or less, 67 ng/ml or less, 66 ng/ml or less, 65 ng/ml or less, 64 ng/ml or less, 63 ng/ml or less, 62 ng/ml or less, 61 ng/ml or less, 60 ng/ml or less, 55 ng/ml or less, 50 ng/ml or less, 45 ng/ml or less, 40 ng/ml or less, 35 ng/ml or less, 34 ng/ml or less, 33 ng/ml or less, 32 ng/ml or less, 31 ng/ml or less, 30 ng/ml or less, 25 ng/ml or less, 20 ng/ml or less, 19 ng/ml or less, 18 ng/ml or less, 17 ng/ml or less, 16 ng/ml or less or 15

ng/ml or less. In one preferred embodiment of the invention, the (2S,6S)-OH-norketamine C.sub.max of said administration is 75 ng/ml or less. In another preferred embodiment of the invention, the (2S,6S)-OH-norketamine C.sub.max of said administration is 35 ng/ml or less. In another preferred embodiment of the invention, the (2S,6S)-OH-norketamine C.sub.max of said administration is 20 ng/ml or less. In one preferred embodiment of the invention, the (2S,6S)-OH-norketamine C.sub.max of said administration is between 15 ng/mL and 75 ng/mL. In another preferred embodiment of the invention, the (2S,6S)-OH-norketamine C.sub.max of said administration is between 15 ng/mL and 20 ng/mL. In another preferred embodiment of the invention, the (2S,6S)-OH-norketamine C.sub.max of said administration is between 30 ng/mL and 35 ng/mL. In another preferred embodiment of the invention, the (2S,6S)-OH-norketamine C.sub.max of said administration is between 60 ng/mL and 75 ng/mL.

[0051] The present invention is further directed to a method of treating major depressive disorder in a human patient in need thereof by the oral administration to said patient of a dosage form, wherein said dosage form provides for an (2S,6S)-OH-norketamine AUC.sub.0-t of 850 ng*h/ml or less.

[0052] In one embodiment of the invention, the (2S,6S)-OH-norketamine AUC.sub.0-t of said administration is 850 ng*h/ml or less, 845 ng*h/ml or less, 840 ng*h/ml or less, 839 ng*h/ml or less, 838 ng*h/ml or less, 837 ng*h/ml or less, 836 ng*h/ml or less, 835 ng*h/ml or less, 834 ng*h/ml or less, 832 ng*h/ml or less, 831 ng*h/ml or less, 830 ng*h/ml or less, 829 ng*h/ml or less, 828 ng*h/ml or less, 827 ng*h/ml or less, 826 ng*h/ml or less, 825 ng*h/ml or less, 824 ng*h/ml or less, 823 ng*h/ml or less, 822 ng*h/ml or less, 821 ng*h/ml or less, 820 ng*h/ml or less, 815 ng*h/ml or less, 810 ng*h/ml or less, 805 ng*h/ml or less, 800 ng*h/ml or less, 795 ng*h/ml or less, 790 ng*h/ml or less, 785 ng*h/ml or less, 780 ng*h/ml or less, 775 ng*h/ml or less, 770 ng*h/ml or less, 765 ng*h/ml or less, 760 ng*h/ml or less, 755 ng*h/ml or less, 750 ng*h/ml or less, 745 ng*h/ml or less, 740 ng*h/ml or less, 735 ng*h/ml or less, 730 ng*h/ml or less, 725 ng*h/ml or less, 720 ng*h/ml or less, 710 ng*h/ml or less, 700 ng*h/ml or less, 690 ng*h/ml or less, 680 ng*h/ml or less, 670 ng*h/ml or less, 660 ng*h/ml or less, 650 ng*h/ml or less, 640 ng*h/ml or less, 630 ng*h/ml or less, 620 ng*h/ml or less, 610 ng*h/ml or less, 600 ng*h/ml or less, 590 ng*h/ml or less, 580 ng*h/ml or less, 570 ng*h/ml or less, 560 ng*h/ml or less, 550 ng*h/ml or less, 540 ng*h/ml or less, 530 ng*h/ml or less, 520 ng*h/ml or less, 510 ng*h/ml or less, 500 ng*h/ml or less, 490 ng*h/ml or less, 480 ng*h/ml or less, 470 ng*h/ml or less, 460 ng*h/ml or less, 450 ng*h/ml or less, 440 ng*h/ml or less, 430 ng*h/ml or less, 425 ng*h/ml or less, 420 ng*h/ml or less, 419 ng*h/ml or less, 418 ng*h/ml or less, 417 ng*h/ml or less, 416 ng*h/ml or less, 415 ng*h/ml or less, 414 ng*h/ml or less, 413 ng*h/ml or less, 412 ng*h/ml or less, 411 ng*h/ml or less, 410 ng*h/ml or less, 405 ng*h/ml or less, 400 ng*h/ml or less, 390 ng*h/ml or less, 380 ng*h/ml or less, 370 ng*h/ml or less, 360 ng*h/ml or less, 350 ng*h/ml or less, 340 ng*h/ml or less, 330 ng*h/ml or less, 320 ng*h/ml or less, 310 ng*h/ml or less, 300 ng*h/ml or less, 290 ng*h/ml or less, 280 ng*h/ml or less, 270 ng*h/ml or less, 260 ng*h/ml or less, 250 ng*h/ml or less, 240 ng*h/ml or less, 230 ng*h/ml or less, 220 ng*h/ml or less, 215 ng*h/ml or less, 210 ng*h/ml or less, 209 ng*h/ml or less, 208 ng*h/ml or less, 207 ng*h/ml or less, 206 ng*h/ml or less, 205 ng*h/ml or less, 200 ng*h/ml or less, 190 ng*h/ml or less, 180 ng*h/ml or less, 170 ng*h/ml or less, 160 ng*h/ml or less, 150 ng*h/ml or less, 140 ng*h/ml or less, 130 ng*h/ml or less, 120 ng*h/ml or less, 110 ng*h/ml or less, 105 ng*h/ml or less, 104 ng*h/ml or less, 103 ng*h/ml or less or 102 ng*h/ml or less. In one preferred embodiment of the invention, the (2S,6S)-OH-norketamine AUC.sub.0-t of said administration is 850 ng*h/ml or less. In another preferred embodiment of the invention, the (2S,6S)-OH-norketamine AUC.sub.0-t of said administration is 420 ng*h/ml or less. In another preferred embodiment of the invention, the (2S,6S)-OH-norketamine AUC.sub.0-t of said administration is 210 ng*h/ml or less. In another preferred embodiment of the invention, the (2S,6S)-OH-norketamine AUC.sub.0-t of said administration is 105 ng*h/ml or less. In one

preferred embodiment of the invention, the (2S,6S)-OH-norketamine AUC.sub.0-t of said administration is between 105 ng*h/ml and 850 ng*h/ml. In another preferred embodiment of the invention, the (2S,6S)-OH-norketamine AUC.sub.0-t of said administration is between 105 ng*h/ml and 850 ng*h/ml. In another preferred embodiment of the invention, the (2S,6S)-OH-norketamine AUC.sub.0-t of said administration is between 102 ng*h/ml and 105 ng*h/ml. In another preferred embodiment of the invention, the (2S,6S)-OH-norketamine AUC.sub.0-t of said administration is between 205 ng*h/ml and 210 ng*h/ml. In another preferred embodiment of the invention, the (2S,6S)-OH-norketamine AUC.sub.0-t of said administration is between 410 ng*h/ml and 420 ng*h/ml. In another preferred embodiment of the invention, the (2S,6S)-OH-norketamine AUC.sub.0-t of said administration is between 820 ng*h/ml and 840 ng*h/ml.

[0053] The present invention is further directed to a method of treating major depressive disorder, in a human patient in need thereof by the oral administration to said patient of an oral dosage form comprising about 5 mg to about 40 mg, for example of esketamine, 5 mg to 40 mg, of esketamine, wherein the administration is daily.

[0054] In one embodiment of the invention, the daily administration of esketamine is provided in a single daily dose. In another embodiment of the invention, the daily administration of esketamine is provided in two doses, in three doses, or in four doses, each dose being spread about equally over the 24 hour period.

[0055] The present invention is further directed to a method of treating major depressive disorder in a human patient in need thereof by the oral administration to said patient of an oral dosage form comprising 5 mg to 40 mg of esketamine, preferably 5 mg to 40 mg of esketamine, wherein the administration is intermittent over the treatment regimen.

[0056] In a preferred embodiment of the invention, the intermittent administration is once every second day to about once a month or once every 4 weeks. In one embodiment of the invention, the intermittent administration is once every second day, once every third day, twice a week, once every fourth day, once every fifth day, once every sixth day, once a week, once every eighth day, once every ninth day, once every tenth day, once every eleventh day, once every twelfth day, once every thirteenth day, once every two weeks, once every three weeks or once a month. In one preferred embodiment of the invention, the intermittent administration is twice a week. In another preferred embodiment of the invention, the intermittent administration is once a week. In yet another preferred embodiment of the invention, the intermittent administration is once a month. In yet another preferred embodiment of the invention, the intermittent administration is once every 4 weeks.

[0057] In one embodiment of the invention, frequency of the intermittent administration can vary over the time period of the treatment regimen. In a preferred embodiment of the invention, the frequency of the intermittent administration is gradually reduced over the time period of the treatment regimen. In a more preferred embodiment of the invention, the frequency of the intermittent administration is reduced from twice a week to once a week. In another preferred embodiment of the invention, the frequency of the intermittent administration is reduced from once a week to once every two weeks. In an even more preferred embodiment of the invention, the frequency of the intermittent administration is reduced from twice a week to once a week to once every two weeks. In another preferred embodiment of the invention, the frequency of the intermittent administration is maintained consistently over the time period of the treatment regimen.

[0058] The present invention is further directed to a method of treating major depressive disorder in a human patient in need thereof by the oral administration to said patient of an oral dosage form comprising about 5 mg to about 40 mg of esketamine, preferably 5 mg to 40 mg of esketamine, over a treatment regimen of at least 28 days wherein the administration is self-administered. As used herein "self-administered" refers to administration wherein the patient is responsible for taking the medication and is not assisted during the oral administration of the oral dosage form by a

healthcare professional. In some aspects, one or more of the administrations may be assisted by a healthcare professional and one or more of the administration may be self-administered over the treatment regimen. In one embodiment, said self-administration is in the patient's own home. In a preferred embodiment, said self-administration is at night. In a more preferred embodiment, said self-administration is before the patient goes to sleep.

[0059] In another embodiment, the patient has no restrictions on driving in the 24 hours immediately following the oral administration of the oral dosage form comprising about 5 mg to about 40 mg of esketamine, preferably 5 mg to 40 mg of esketamine. That is, the oral administration of the oral dosage form does not result in a mental or motor impairment that negatively affects the patient's ability to operate a motor vehicle. In the 24 hours immediately following the administration.

[0060] In yet another embodiment, the patient is restricted from driving for no more than 10 hours after the oral administration of the oral dosage form comprising about 5 mg to about 40 mg of esketamine, preferably 5 mg to 40 mg of esketamine. In a preferred embodiment, the patient is restricted from driving for no more than 8 hours after the administration. In another preferred embodiment, the patient is restricted from driving for no more than 6 hours after the administration. In a more preferred embodiment, the patient is restricted from driving for no more than 2 hours after the administration. In a most preferred embodiment, the patient is restricted from driving for no more than an hour after the oral administration of the oral dosage form comprising about 5 mg to about 40 mg of esketamine, preferably 5 mg to 40 mg of esketamine.

[0061] In one embodiment of the invention, the oral dosage form is a liquid preparation such as a suspension, elixir, or solution. In another embodiment of the invention, the oral dosage forms are solid preparations, for example, powders, capsules, caplets, gelcaps, and tablets. In a preferred embodiment, the oral dosage form is a tablet, gelcap, or capsule. In a more preferred embodiment, the oral dosage form is a tablet.

[0062] To prepare the preparations, i.e., the oral dosage forms, of this invention, esketamine, and optionally, at least one second medication other than (R)-ketamine, are admixed with pharmaceutical carriers according to conventional pharmaceutical compounding techniques, which carriers may take a wide variety of forms depending of the form of preparation desired for administration. In preparing the oral preparations, any of the usual pharmaceutical media may be employed. Thus, for liquid oral preparations, such as for example, suspensions, elixirs and solutions, suitable carriers and additives include water, glycols, oils, alcohols, flavoring agents, preservatives, coloring agents and the like. For solid oral preparations such as, for example, powders, capsules, caplets, gelcaps and tablets, suitable carriers and additives include starches, sugars, diluents, granulating agents, lubricants, binders, disintegrating agents and the like. Because of their ease in administration, tablets and capsules represent the most advantageous oral dosage unit form, in which case solid pharmaceutical carriers are employed.

[0063] The present invention is further directed to a method of treating major depressive disorder in a human patient in need thereof by the administration to said patient of an oral dosage form comprising about 5 mg to about 40 mg of esketamine, preferably 5 mg to 40 mg of esketamine, wherein the oral dosage form is an abuse deterrent formulation. In a more preferred embodiment, the abuse deterrent formulation is a tablet. Abuse deterrent tablet formulations can be prepared by methods known in the art including as found in U.S. Pat. No. 7,955,619, WO2014006004, WO2008033523, WO2008023261, WO2016094358 and US2004052731 each of which is hereby incorporated by reference.

[0064] The present invention is further directed to methods of treating major depressive disorder in a human patient in need thereof by the oral administration to said patient of an oral dosage form comprising about 5 mg to about 40 mg of esketamine, preferably 5 mg to 40 mg of esketamine, further comprising the administration of a second medication other than (R)-ketamine.

[0065] In a preferred embodiment, the second medication is an antidepressant, an antimanic agent

or an anxiolytic drug. In one embodiment of the invention, the antidepressant is selected from the group consisting of mono-amine oxidase inhibitors (MAOI), tricyclic antidepressants (TCA), serotonin specific reuptake inhibitors (SSRI), serotonin noradrenergic reuptake inhibitors (SNRI), noradrenaline reuptake inhibitor (NRI), "natural products" (such as Kava-Kava, St. John's Wort), dietary supplement (such as s-adenosylmethionine) and others. More specifically, antidepressants include, but are not limited to, imipramine, amitriptyline, desipramine, nortriptyline, doxepin, protriptyline, trimipramine, maprotiline, amoxapine, trazodone, bupropion, chlomipramine, fluoxetine, citalopram, escitalopram, sertraline, paroxetine, tianeptine, agomelatine, nefazadone, venlafaxine, desvenlafaxine, vilazodone, vortioxetine, duloxetine, reboxetine, mirtazapine, mianserin, phenelzine, tranylcypromine, and/or moclobemide.

[0066] In another preferred embodiment, the second medication is an antimanic agent. In one embodiment of the invention, the antimanic agent is selected from the group consisting of carbamazepine, gabapentin, Lithium or a pharmaceutically acceptable salt thereof, valproic acid and antipsychotic medications such as lurasidone, cariprazine, olanzapine, risperidone, quetiapine, paliperidone, aripiprazole and brexpiprazole.

[0067] In another preferred embodiment, the second medication is an anxiolytic drug. In one embodiment of the invention, the anxiolytic drug is selected from the group consisting of Alprazolam, Bromazepam, Chlordiazepoxide, Clonazepam, Clorazepate, Diazepam, Flurazepam, Lorazepam, Oxazepam, Temazepam, Triazolam, Buspirone, Gepirone, Ispapirone, Hydroxyzine, Amobarbital, Pentobarbital, Phenobarbital, Thiopental and Propanolol.

[0068] The present invention is further directed to a method of treating major depressive disorder in a human patient in need thereof, by the oral administration to said patient of an oral dosage form comprising about 5 mg to about 40 mg of esketamine, preferably 5 mg to 40 mg of esketamine and an antidepressant selected from the group consisting of fluoxetine, sertraline, paroxetine, citalopram, escitalopram, venlafaxine, desvenlafaxine, duloxetine, and fluvoxamine and wherein the administration is daily over a treatment regimen of at least 28 days.

[0069] This invention will be better understood by reference to the Examples, which follow, but those skilled in the art will readily appreciate that the specific experiments detailed are only illustrative of the invention as described more fully in the claims which follow thereafter.

EXAMPLES

Example 1: In Vitro Chromosomal Aberration Assay

Example 1a: Esketamine

[0070] The clastogenic potential of esketamine was evaluated in the in vitro mammalian chromosome aberration test using human peripheral blood lymphocytes (HPBL) in both the absence and presence of an induced metabolic activation system (the 9000 g supernatant [S9] microsomal fraction of liver homogenate from rats exposed to Aroclor-1254). Clastogenicity was evaluated by microscopic examination of HPBL in metaphase to determine the mitotic index (MI) and percentage of metaphase cells with numerical and/or structural chromosome aberrations.

[0071] The study was conducted in 2 phases, with a preliminary toxicity test used to determine the appropriate concentrations for the definitive chromosomal aberration assay. Water was used as a negative (vehicle) control. Toxicity (defined as $\geq 450\%$ reduction in MI relative to the vehicle control) was evaluated at 9 concentrations, ranging from 0.0238 to 238 g/ml, after exposure of HPBL to esketamine for 20 hours in the absence of S9 activation, or for 4 hours, either in the presence or absence of S9 activation, followed by a 16-hour recovery period. Toxicity was not observed at any dose in any of the three treatment conditions. Based upon these results, the doses chosen for the chromosomal aberration assay ranged from 30 to 238 g/ml for all three treatment conditions. All concentrations were between 98% to 101% of the nominal concentrations.

[0072] The definitive chromosome aberration assay evaluated HPBL cells after exposure to esketamine for 20 hours in the absence of S9 activation, or for 4 hours, either in the presence or absence of S9 activation, followed by a 16-hour recovery. Positive controls for chromosomal

aberrations in non-activated and S9-activated evaluations, respectively, consisted of mitomycin C (MMC, 0.6 and 0.3 g/mL for the 4- and 20-hour exposures, respectively) and cyclophosphamide (CP, 2.5, 5, and 7.5 g/mL). Water was associated with mean MI values from 13.4% to 16.5% and numerical or structural chromosomal aberrations in 0% to 0.7% of the cells. In non-activated systems, MMC was associated with a mean MI value of 9% with structural chromosomal aberrations in 13.3% of the cells. In the S9-activated system, CP was associated with a mean MI value of 6.2% and with structural aberrations in 10.7% of the cells. The results for negative controls were within the range of historical controls and the results for positive controls were statistically significant ($p < 0.01$, Fisher's exact test). Thus the requirements for a valid test were fulfilled. In the chromosomal aberration assay, cytotoxicity ($\geq 45\%$ reduction in mitotic index relative to the vehicle control) was not observed at any esketamine dose in the non-activated 4- and 20-hour treatment conditions. Cytotoxicity was observed at doses ≥ 200 g/mL in the S9-activated 4-hour exposure group. Initially, the doses selected for evaluation of chromosomal aberrations were 60, 120, and 238 g/mL for the non-activated 4- and 20-hour treatment conditions; and 30, 60, and 200 g/mL for the S9-activated 4-hour treatment condition.

[0073] In the non-activated 4 and 20-hour exposure groups, no significant or dose-dependent increases in structural or numerical (polyploid or endoreduplicated cells) aberrations were observed at any dose ($p > 0.05$; Fisher's Exact and Cochran-Armitage tests).

[0074] In the S9-activated 4-hour exposure group, a statistically significant increase (5.0%) in structural aberrations was observed at 200 g/mL ($p \leq 0.01$; Fisher's Exact test). In order to confirm that the statistical significance observed at the high dose was not due to cytotoxicity, a lower dose (120 g/mL) was included in the evaluation. A statistically significant increase (4.3%) in structural aberrations was observed at 120 g/mL ($p < 0.01$; Fisher's Exact test). The Cochran-Armitage test was positive for a dose response ($p \leq 0.01$). No significant or dose-dependent increases in numerical (polyploid or endoreduplicated cells) aberrations were observed at any dose ($p > 0.05$; Fisher's Exact and Cochran-Armitage tests).

[0075] The results of the study indicate that esketamine was positive for the induction of structural chromosomal aberrations and negative for the induction of numerical chromosomal aberrations in the presence of the exogenous metabolic activation system. Esketamine was negative for the induction of structural and numerical chromosomal aberrations in the absence of the exogenous metabolic activation system.

Example 1b: (S)-Norketamine

[0076] The clastogenic potential of (S)-norketamine was evaluated in the in vitro mammalian chromosome aberration test using human peripheral blood lymphocytes (HPBL) in both the absence and presence of an induced metabolic activation system (the 9000 g supernatant [S9] microsomal fraction of liver homogenate from rats exposed to phenobarbital/5,6-benzoflavone). Clastogenicity was evaluated by microscopic examination of HPBL in metaphase to determine the percentage of metaphase cells with numerical and/or structural chromosome aberrations.

[0077] The study was conducted in 2 phases, with a preliminary toxicity test used to determine the appropriate concentrations for the definitive chromosomal aberration assay. Water was used as a negative (vehicle) control. In both phases, the cells were treated for 3 and 21 hours in the absence of S9 mix and for 3 hours in the presence of S9 mix. The mitotic index was assessed for all cultures to determine cytotoxicity. Ten concentrations, ranging from 2.62 to 260.16 $\mu\text{g/mL}$, were evaluated in the preliminary toxicity test. Toxicity was not observed at any dose in any of the three treatment conditions. Based on these results, the highest concentration for the definitive chromosomal aberration assay was based on the limit concentration (260.16 $\mu\text{g/mL}$, 1 mM) for this test system, where relatively no cytotoxicity was observed. (S)-Norketamine concentrations of 93.66, 156.10 or 260.16 $\mu\text{g/mL}$ were selected for metaphase analysis.

[0078] (S)-Norketamine caused no statistically significant increases in the proportion of metaphase figures containing chromosomal aberrations, at any analyzed concentration, when compared with

the vehicle control. All mean values for the vehicle control (water), and all (S)-norketamine treatment concentrations were below to the laboratory historical control range, when taken at the upper 95% control limit.

[0079] No statistically significant increases in the proportion of polyploid or endoreduplicated metaphase cells were observed during metaphase analysis, under any treatment condition, when compared with the vehicle control whereas both positive control compounds, mytomicin C and cyclophosphamide, caused statistically significant increases in the proportion of aberrant cells, demonstrating the sensitivity of the test system and the efficacy of the S9 mix.

[0080] In conclusion, the results of the in vitro mammalian chromosome aberration test using human peripheral blood lymphocytes indicate that (S)-norketamine has shown no evidence of causing an increase in the frequency of structural chromosome aberrations with or without S9. Therefore, under the conditions of this experiment, (S)-norketamine was non-clastogenic, or negative for the induction of structural and numerical chromosomal aberrations.

Example 2: In Vivo Single Cell Gel Electrophoresis Assay and Mammalian Erythrocyte Micronucleus Test of Esketamine in Sprague Dawley Rats

[0081] The potential of esketamine to induce DNA strand breaks in the liver and also assess the potential induction of micronuclei in the bone marrow cells of Crl:CD(SD) rats. Animals were treated with esketamine orally on three occasions, the second dose being administered approximately 24 hours after the first dose, with the third dose being administered approximately 21 hours after the second dose, 3 hours before sampling. All animals were dosed orally by gavage using a dose volume of 10 mL/kg.

[0082] Substantial differences in toxicity were observed between the sexes in the preliminary toxicity test, therefore, in line with current guidelines the test was performed using both male and female animals. Dose levels of 18.75, 37.5 and 75 mg/kg/day (male animals) and 12.5, 25 and 50 mg/kg/day (female animals) were selected. The vehicle control group received purified water and the positive control group for the comet phase received Ethyl Methanesulfonate at 200 mg/kg. Blood samples were taken via the tail vein on Day 3 prior to dosing, at 30 minutes and 3 hours post dose from satellite animals and all main study animals prior to termination.

[0083] Cell suspensions from each tissue were obtained from animals in the vehicle control group and in each of the test item groups approximately 3 hours after administration of the third dose. Cell suspensions from animals in the positive control group were obtained approximately 3 hours after a single dose.

[0084] Following electrophoresis three slides per animal per tissue were analyzed for comets. Slides were visualized by staining with SYBR GOLD® via fluorescence microscopy. 150 morphologically normal cells were analyzed for the presence of comets per animal per tissue. DNA strand breaks were assessed by comparing the mean and median % tail intensities (% TI) from esketamine treated animals compared to vehicle control values. The slides were also examined for any overt toxicity, e.g. an increase in background debris and/or an increase in the incidence of excessively damaged cells (i.e. Hedgehog cells). These cells were excluded from the analysis, along with any cells that had unusual staining artefacts.

[0085] Bone marrow smears were obtained from animals in the vehicle control and in each of the test item groups approximately 3 hours after administration of the third dose. In addition, slides prepared from a separate study [CT12GD] from animals treated with Cyclophosphamide a well characterized clastogen, were stained and coded along with the bone marrow smears prepared from animals in this study.

[0086] One smear from each animal was examined for the presence of micronuclei in 4000 polychromatic erythrocytes. The proportion of polychromatic erythrocytes was assessed by examination of at least 1000 erythrocytes from each animal. A record of the incidence of micronucleated normochromatic erythrocytes was also kept.

[0087] Statistically significant increases in the median % (TI) were observed in the liver of male

Crl:CD(SD) rats administered esketamine at 75 mg/kg/day ($p<0.001$) compared to vehicle control values. The group mean and median % TI values for male animals administered esketamine at 75 mg/kg/day were outside of the current vehicle historical control range. Statistically significant increases in the median % TI were observed in the liver of female Crl:CD(SD) rats administered esketamine at 25 and 50 mg/kg/day ($p<0.001$) compared to vehicle control values. The group mean and median % TI values for female animals administered esketamine at 25 and 50 mg/kg/day were outside of the current vehicle historical control range.

[0088] The positive control compound, Ethyl methanesulphonate, produced significant increases in the median % TI when compared to vehicle control values in male and female animals ($p<0.001$, t-test). No Hedgehog cells were observed in the liver of male or female Crl:CD(SD) rats administered esketamine at any dose level, compared to vehicle control values.

[0089] Sections of the liver from the vehicle control animals and animals administered esketamine at 75 mg/kg/day (male animals) and 25 and 50 mg/kg/day (female animals) were processed for histopathological examination and assessed for signs of cytotoxicity, necrosis and apoptosis. Increased hepatocellular mitotic figures were observed some males animals given 75 mg/kg/day. The macroscopic examination performed after 3 doses of treatment revealed no test item related lesions.

[0090] No statistically significant increases in the frequency of micronucleated polychromatic erythrocytes were observed in male Crl:CD(SD) rats administered esketamine at any dose level compared to vehicle control values. All individual and group mean values were within the current vehicle historical control range (control limits).

[0091] Statistically significant decreases in the proportion of polychromatic erythrocytes were observed in male Crl:CD(SD) rats administered esketamine at 37.5 mg/kg/day (pairwise and trend test, $p<0.05$) and 75 mg/kg/day (trend test, $p<0.05$), compared to vehicle control values. All individual and group mean values were within the current vehicle historical control range (control limits); therefore this result is not considered to be biologically relevant.

[0092] No statistically significant increases in the frequency of micronucleated polychromatic erythrocytes and no statistically significant decreases in the proportion of polychromatic erythrocytes were observed in female Crl:CD(SD) rats administered esketamine at any dose level, compared to vehicle control values. All individual and group mean values were within the current vehicle historical control range (control limits). In accordance with ICH S2(R1) the coded positive control slides prepared from the study CT12GD demonstrated the ability of the scorer to detect increases in micronucleated polychromatic erythrocytes.

[0093] The results of the study indicate that esketamine has shown evidence of causing an increase in DNA strand breaks in the liver of male and female Crl:CD(SD) rats when administered orally by gavage but has not shown any evidence of causing an increase in the induction of micronucleated polychromatic erythrocytes or bone marrow cell toxicity in male or female Crl:CD(SD) rats when administered orally by gavage.

[0094] Using PROAST v63.3 (in development), the benchmark dose (BMD50) was modelled based on the mean and median tail intensity values respectively, for the male and female rats following exposure to esketamine. The Hill and exponential models provided a suitable fit to the in vivo comet tail intensity data, which is consistent with the non-linear dose response. The lower benchmark dose (BMDL50) metrics were calculated to be 9.83 mg/kg/day in female rats and 27.31 mg/kg/day in male rats, both using the 'single slide median Tail Intensity' which were lower and more conservative than those derived when using the 'single slide mean Tail Intensity'. These point of departure (POD) metrics are comparable to the no observed genotoxic effect level for comet tail intensity in liver at 12.5 mg/kg/day for female and 37.50 mg/kg/day for male rats.

Example 3: Repeated Dose 28 Day Toxicokinetic Study of Esketamine in Sprague Dawley Rats

[0095] The objective of the study was to assess the potential toxicity, neurobehavioral effects, and toxicokinetics (TK) of esketamine when administered orally, via gavage, to Sprague Dawley rats

for 28 days and to evaluate recovery during a 14-day drug-free period. Fifty male and 50 female rats were randomized into 4 groups (15/sex/Groups 1 and 4; 10/sex/Groups 2 and 3). Esketamine was administered via oral gavage once daily for 28 consecutive days to males at 0 (vehicle control), 6, 10 or 30 mg/kg/day and females at 0 (vehicle control), 2, 10 or 20 mg/kg/day in a dose volume of 10 mL/kg. Animals were observed until euthanized and necropsied on Day 29 (10/sex/group) or 43 (5/sex from Groups 1 and 4). Toxicity was evaluated based on mortality, clinical observations, body weights, food consumption, ophthalmology, motor activity, functional observational battery, clinical pathology (clinical chemistry, hematology, coagulation and urinalysis), organ weights, anatomic (macroscopic or microscopic) pathology. Toxicokinetic animals (3/sex/Group 1; 6/sex/Groups 2, 3, and 4) were similarly dosed and bled on Day 1 and during Week 4 for toxicokinetic analysis.

[0096] There was no mortality found in this study and there were no esketamine-related effects on clinical signs, body weights, food consumption, ophthalmology, motor activity, functional observational battery, clinical pathology or anatomic pathology changes.

[0097] Esketamine exposure increased in a generally dose-proportional manner in males and in a slightly greater than dose-proportional manner in females over the dose ranges of 6 to 30 mg/kg/day for males and 2 to 20 mg/kg/day for females. After normalization for dose level differences, males had lower exposures than females. Exposures were similar on Day 28 compared to Day 1, with the exception of C_{max} in females, which was higher on Day 28. The results of the esketamine exposure at day 1 are described in Table 1, and at day 28 in Table 2.

TABLE-US-00001																													
TABLE 1 Dose normalized Esketamine AUC.sub.0-t dose t.sub.max C.sub.max																													
AUC.sub.0-t AUC.sub.0-inf (ng*h/ml)/ t.sub.1/2 Sex (mg/kg) (h) (ng/ml) (ng*h/ml) (ng*h/ml)																													
(mg/kg) (h) F 2 0.17 67.2 57 58 29.0 0.5 F 10 0.17 294.9 418 440 44.0 0.7 F 20 0.17 671.8 993 995																													
49.8 0.9 M 6 0.17 108.7 99 103 17.2 0.8 M 10 0.17 179.2 175 181 18.1 0.7 M 30 0.17 296.0 580																													
585 19.5 1.2																													

TABLE-US-00002		TABLE 2		Dose normalized Esketamine AUC.sub.0-t dose t.sub.max C.sub.max		AUC.sub.0-t (ng*h/ml)/ t.sub.1/2		Sex (mg/kg) (h) (ng/ml) (ng*h/ml) (mg/kg) (h)		F		2		0.17		132.7		62		31.0		NC		F		10		0.17		767.9		514		51.3		0.4		F		20		0.17		1451.1		1064		53.2		0.6		M		6		0.17		113.1		61		10.2		0.6		M		10		0.17		211.3		179		17.9		0.6		M		30		0.5		398.0		575		19.2		1.5	
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Example 4: Long Term Carcinogenicity Study

[0098] A 104 Week carcinogenicity study of esketamine administered via oral gavage to Sprague Dawley Rats is performed to evaluate the carcinogenic potential and determine the toxicokinetics of esketamine.

[0099] As based on the International Conference on Harmonization (ICH) S1 Guidelines S1A, Guideline on the Need for Carcinogenicity Studies of Pharmaceuticals; SIB, Testing for Carcinogenicity of Pharmaceuticals; and S1C(R2), Dose Selection for Carcinogenicity Studies of Pharmaceuticals, 236 male and 236 female Sprague Dawley Rats are administered esketamine over 104 weeks at the doses of 0 (vehicle control), 6, 10 or 30 mg/kg/day for the male rats and 0 (vehicle control), 2, 10 or 20 mg/kg/day for the female rats.

[0100] The study end-points include clinical observations, body weight changes, food consumption, bioanalytical toxicokinetic analysis, and anatomic macroscopic and microscopic pathology findings.

[0101] It can thus be demonstrated that the genotoxic changes as shown in Examples 1 and 2 were not identified after 28 days administration at point of departure doses and at reduced doses, which factor in an at least 10 fold safety margin after 730 days, thereby providing a minimal safe window for chronic esketamine administration.

Example 5: 7-Day Forced Swim Test in Male Rates

[0102] Groups of, 6-7 week old, male Sprague Dawley rats were administered esketamine by intraperitoneal injection and their behavioral despair assessed by a forced swim test. The animals, in cohorts of 10, were administered either a single dose of 15 mg/kg esketamine, 7 daily doses of

either 7.5 or 15 mg/kg esketamine or a vehicle control, and the test performed 30 minutes after dosing. Statistical evaluation was performed using an ordinary One-Way ANOVA, t Test and an Uncorrected Fisher's LSD comparisons test.

[0103] Continuous 7-day treatment of esketamine produced stronger antidepressant-like effect than a single acute dose at the same doses levels. Esketamine at 7.5 and 15 mg/kg, exerted a statistically significant decrease of 40% and 60% respectively in immobility time following chronic treatment, whereas the extent of the effect was less marked in acute treated rats (42% of control for 15 mg/kg). This indicates that multiple dosing is more effective than a single acute dose of the same dose level and suggests a rationale to treat depressed patients via a chronic, rather than acute, esketamine regimen.

Example 6: A Single Dose, Randomized, Open-Label, Crossover Study in Healthy Volunteers

[0104] A randomized, open-label 4-way crossover study in 16 healthy male and female subjects was held wherein said subjects were placed in a randomly assigned order and administered esketamine. Each subject was assigned to 1 of 4 treatment sequences according to a randomization code such that 4 subjects were assigned to each treatment sequence. There was a wash-out period of at least 7 days between dosing periods with doses consisting of either oral or intravenous esketamine hydrochloride. The study consisted of an eligibility screening period of 28 days, 4 study periods involving administration of a single dose of esketamine hydrochloride followed by safety assessments with blood sampling for PK purposes up to 72 hours after study drug administration, discharge at 72 hours after study drug administration and a follow-up visit 7-14 days after the last PK blood sample was taken on Day 4.

[0105] Fifteen of 16 subjects completed the study. One subject (Subject 11) participated in the first treatment period only. This subject was withdrawn from the study due to an AE of mild hyperbilirubinemia and therefore did not receive the planned treatments in the 3 remaining treatment periods. Subject 11 was not included in the PK set, as presented in Table 3, which therefore included 15 subjects.

TABLE-US-00003 TABLE 3

	20 mg oral solution	20 mg oral tablet	100 mg oral kg iv	0.3 mg/ Esketamine	Parameter	Statistic	tablet
C.sub.max (ng/ml)	11.92	16.48	65.34	94.50	Geometric mean		
(CV %)	(45%)	(48%)	(44%)	(33%)			
t.sub.max (h)	0.75	0.50	0.75	1.00	Median		
AUC.sub.0-t	20.05	23.44	172.71	249.15	Geometric mean		
(CV %)	(21%)	(19%)	(19%)	(18%)			
AUC.sub.0-inf	22.46	25.82	180.84	257.14	Geometric mean		
(CV %)	(58%)	(39%)	(48%)	(21%)			
t.sub.1/2 (h)	2.84	2.91	7.66	10.16	Geometric mean		
(CV %)	(37%)	(33%)	(41%)	(26%)			
(S)-norketamine C.sub.max (ng/ml)	89.73	99.78	351.89	42.25	Geometric mean		
(CV %)	(22%)	(26%)	(24%)	(16%)			
t.sub.max (h)	1.00	0.75	1.00	1.25	Median		
AUC.sub.0-t	418.33	404.96	2267.52	425.96	Geometric mean		
(CV %)	(22%)	(19%)	(19%)	(18%)			
AUC.sub.0-inf	429.05	416.89	2282.91	440.76	Geometric mean		
(CV %)	(22%)	(19%)	(19%)	(18%)			
t.sub.1/2 (h)	8.75	8.88	9.27	11.26	Geometric mean		
(CV %)	(28%)	(26%)	(25%)	(28%)			
(2S,6S)-OH-Norketamine C.sub.max (ng/ml)	45.75	46.46	189.32	24.79	Geometric mean		
(CV %)	(33%)	(28%)	(24%)	(28%)			
t.sub.max (h)	1.50	1.00	2.00	3.00	Median		
AUC.sub.0-t	390.10	365.99	1945.85	376.15	Geometric mean		
(CV %)	(27%)	(32%)	(18%)	(28%)			
AUC.sub.0-inf	400.91	377.68	1959.73	389.50	Geometric mean		
(CV %)	(26%)	(31%)	(18%)	(27%)			
t.sub.1/2 (h)	7.77	7.83	8.92	10.68	Geometric mean		
(CV %)	(31%)	(30%)	(26%)	(25%)			

[0106] Amongst the 15 subjects who completed the study, the treatment was generally well tolerated. A total of 128 TEAEs (treatment-emergent adverse event) were reported by 15 of 16 (94%) subjects of which 79 TEAEs reported by 14 of 16 (88%) subjects were to be related to the study drug. Overall, a total of 14 of 128 TEAEs reported by 4 (25%) subjects were of moderate severity and 114 of 128 TEAEs reported by 15 (94%) subjects were of mild severity. No severe TEAEs or SAEs were reported. The most frequently occurring adverse events (reported more than twice) were headache, dizziness, hypokinesia, feeling abnormal, fatigue, euphoric mood

inappropriate affect, nausea and hyperhidrosis. Table 4 presents the frequency of the most frequently reported related TEAEs as a percentage of the subjects that experienced an adverse effect per treatment.

TABLE-US-00004 TABLE 4 20 mg oral 20 mg oral 100 mg oral 0.3 mg/ TEAE tablet/%
solution/% tablet/% kg iv/% Headache — 19 13 19 Dizziness 13 — 13 6 Hypokinesia — — 13 6
Feeling abnormal 7 19 20 13 Fatigue 7 — — 6 Euphoric mood — — 13 19 Inappropriate — 6 7 6
affect Nausea — — 20 6 Hyperhidrosis 7 — 7 —

Example 7: Oral Esketamine Dosage Forms

[0107] Oral dosage forms of esketamine hydrochloride were manufactured according to the procedures described in WO2016094358, which is incorporated in its entirety, herein, by reference.

[0108] A population PK modeling analyses was performed on the results of Example 5 to provide simulated PK data following a multiple dose regimen of the doses described herein.

[0109] The formulations and their predicted steady state PK parameters are presented in Table 5.

TABLE-US-00005 TABLE 5 5 mg 10 mg 20 mg 40 mg tablet* tablet tablet tablet Ingredient (%
(w/w)) 16.0% Coated Esketamine 312.5 6.25 12.50 25.00 Granules Coated Polymer Granules
30.55 27.42 21.17 7.06 Mannitol 28.13 28.13 28.13 29.74 Crospovidone 20.00 20.00 20.00 20.00
Microcrystalline Cellulose 13.00 13.00 13.00 13.00 Carbomer 2.00 2.00 2.00 2.00 Sodium
Bicarbonate 2.00 2.00 2.00 2.00 Colloidal silicon Dioxide 0.20 0.20 0.20 0.20 Magnesium stearate
1.00 1.00 1.00 1.00 Predicted steady state pharmacokinetic parameters Esketamine C.sub.max
(ng/ml) 2.04 4.09 9.03 18.33 Esketamine AUC.sub.0-tau (ng .Math. h/ml) 6.4 12.7 29.0 61.8 (S)-
norketamine C.sub.max (ng/ml) 16.46 32.92 68.63 136.02 (S)-norketamine AUC.sub.0-tau(ng
.Math. h/ 97.3 194.6 413.4 835.2 ml) *data calculated based on 10 mg predicted data

Example 7: A Dose Range Finding, Multicenter. Double-Randomized, Double-Blind, Placebo-
Controlled Phase II Study, to Determine the Safety and Efficacy 10 mg, 20 mg and 40 mg Oral
Esketamine as an Adjunctive Therapy in Major Depressive Disorder (MDD) Patients with an
Inadequate Response to Standard Anti-Depressant Therapy

[0110] A phase II, dose range finding, multinational, double-randomized, double-blind, placebo-
controlled study compares the efficacy, safety and tolerability of once daily 10, 20 or 40 mg oral
esketamine to placebo treatment in 204 MDD subjects with inadequate response to antidepressant
therapy. All subjects remain on their current anti-depressant with no dose change during the study.

[0111] The study comprises 3 phases, screening (Days 0-28), double-blind treatment (days 29-56)
composed of two 2-week periods (period 1, period 2) and post-treatment safety follow-up (days 57-
70) following the last study treatment administration.

[0112] During screening, subjects are assessed for study eligibility and washed out from disallowed
drugs. After being found eligible, subjects are randomized at the beginning of Period 1 using a
3:1:1:1 allocation scheme to receive, once daily, either placebo or 10 mg, 20 mg or 40 mg oral
esketamine, respectively. At the conclusion of period 1, all subjects are blindly assessed for
response based on their change in MADRS-10 score from baseline to week 2. Subjects who
received placebo during period 1 are re-randomized using a 1:1:1:1 allocation scheme to receive in
the 2 weeks (period 2) either placebo or 10 mg, 20 mg or 40 mg oral esketamine once daily,
respectively. The re-randomization is stratified by the placebo response in Period 1 (Change in
MADRS < or ≥50% and MADRS score < or ≥18). Subjects that were on oral esketamine in period
1 remain on the drug at the same dosage for the 2 weeks of period 2.

[0113] Subjects receive the first dose of their study drug at the study site and are then closely
monitored for 3 hours to assess for potential neuropsychiatric adverse events using a
comprehensive set of scales to identify sleepiness, sedation and dissociative effects. Thereafter, the
subject is provided with a 1 week supply of the study drug for administration at their place of
residence and instructed to take the study drug in the evening (except at the day of the weekly visit
when it is taken at the study site) and not to drive until the next morning. At every subject visit, a
psychiatrist evaluates the subject's MADRS-10 score.

[0114] The study's primary efficacy endpoint is the change from baseline to week 2 (in the 2 periods) in 10 items Montgomery-Asberg Depression Rating Scale (MADRS-10).

[0115] Secondary efficacy endpoints include the change from baseline Sheehan disability scale (SDS) at 2 weeks, remission rate at 2 weeks (MADRS-10 $\leq$ 10), responder rate at 2 weeks ($\geq$ 50% improvement in MADRS-10), change from baseline in self-rated Symptoms of Depression Questionnaire (SDQ) at 2 weeks, physician administered Clinical Global Impression Improvement (CGI-I) at 2 weeks and the change from baseline in Physician administered Clinical Global Impression Severity (CGI-S) at 2 weeks.

[0116] Exploratory Endpoints include the change from baseline in Generalized Anxiety Disorder 7 items scale (GAD-7) at 2 weeks and the change from baseline to week 4 in Montgomery-Asberg Depression Rating Scale (MADRS-10) for the subset of subjects receiving the same study drug for the 4 weeks.

[0117] Safety and tolerability endpoints include adverse events, hematology, biochemistry and urinalysis, immunological parameters, physical examination findings, blood pressure and heart rate every 30 minutes for the 3 hours following study drug administration, 12 lead ECG findings, withdrawal rates, Digit Symbol Substitution Test (DSST), reaction time test (Cambridge COGNITION), self-administered Stanford sleepiness scale, a Bladder Pain/Interstitial Cystitis Symptom Score (BPIC-SS), a Modified Observer's Alertness/Sedation Scale (MOAA/S), a Clinician-Administered Dissociative States Scale (CADSS), a suicidality scale-Clinician-Rated Columbia Suicide Severity Rating Scale (C-SSRS), 4 items positive symptoms subscale from the Brief Psychiatric Rating Scale (BPRS) and 20 item Physician Withdrawal Checklist (PWC-20) during the follow-up period.

[0118] Those skilled in the art will appreciate that numerous changes and modifications can be made to the preferred embodiments of the disclosure and that such changes and modifications can be made without departing from the spirit of the disclosure. It is, therefore, intended that the appended claims cover all such equivalent variations as fall within the true spirit and scope of the disclosure.

Claims

1. A method of treating major depressive disorder (MDD) in a human patient in need thereof comprising orally administering to said patient an oral dosage form consisting of esketamine and a pharmaceutical carrier over a treatment regimen of at least 28 days and wherein the ratio of S-norketamine C.sub.max to esketamine C.sub.max after the administration of the esketamine is at least 6.
2. The method of claim 1, wherein the treatment regimen is between 28 days and about 730 days.
3. The method of claim 2, wherein the treatment regimen is for at least a year.
4. The method of claim 1, wherein the administration of the esketamine is at least once daily over the treatment regimen.
5. The method of claim 4, wherein the administration of the esketamine is once daily over the treatment regimen.
6. The method of claim 1, wherein the administration of the esketamine is intermittent over the treatment regimen.
7. The method of claim 6, wherein the frequency of the administration of the esketamine varies over the treatment regimen.
8. The method of claim 1, wherein the esketamine is esketamine hydrochloride.
9. The method of claim 1, wherein the treatment comprises a reduction in the human patient's symptoms of MDD.
10. The method of claim 9, wherein the reduction in the human patient's symptoms of MDD is measured, as compared to baseline, at at least day 28, as measured by a reduction on the

Montgomery-Asberg Depression Rating Scale.

11. The method of claim 9, wherein the reduction in the human patient's symptoms of MDD is measured, as compared to baseline, at at least day 28, on a Clinical Global Impression Improvement score.

12. The method of claim 9, wherein the reduction in the human patient's symptoms of MDD is measured, as compared to baseline, at at least day 28, on a Global Impression Severity score.

13. The method of claim 9, wherein the human patient's symptoms of MDD are reduced at at least day 28 as compared to placebo.

14. The method of claim 1, further comprising the administration of a second medication other than (R)-ketamine and wherein the ratio of S-norketamine C.sub.max to esketamine C.sub.max after the coadministration of the esketamine and the second medication is at least 6.

15. The method of claim 14, wherein the second medication is an antidepressant, an antimanic agent, or an anxiolytic drug.

16. The method of claim 14, wherein the antidepressant is selected from the group consisting of fluoxetine, sertraline, paroxetine, citalopram, escitalopram, venlafaxine, desvenlafaxine, duloxetine, and fluvoxamine.

17. A method of treating major depressive disorder (MDD) in a human patient in need thereof comprising orally administering to said patient an oral dosage form consisting of esketamine and a pharmaceutical carrier over a treatment regimen of at least 28 days, wherein the ratio of S-norketamine C.sub.max to esketamine C.sub.max after the administration of the esketamine is at least 6, and wherein the administration results in a reduction in the human patient's symptoms of MDD, when measured at at least day 28 as compared to placebo.

18. The method of claim 17, further comprising the administration of a second medication other than (R)-ketamine and wherein the ratio of S-norketamine C.sub.max to esketamine C.sub.max after the coadministration of the esketamine and the second medication is at least 6.

19. The method of claim 1, wherein the second medication is administered to the patient.

20. The method of claim 18, wherein the second medication is administered to the patient.
